CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in generative AI and LLM-based applications, from RAG systems and agents to MCP servers and answer-quality evaluation. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in machine learning and Python development.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Sole developer owning an LLM-based knowledge-extraction pipeline end-to-end, from concept and prompt engineering to data ingestion, chunking, and indexing, reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized a multi-agent system with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for regulatory and technical documents, evaluated against an SAP-system-derived ground truth; drove precision/recall from 33%/45% to 93%/90%.
- Designed and implemented a Model Context Protocol (MCP) server exposing the knowledge graph for millisecond-latency retrieval, integrated into Claude Code; used Claude Code as standard practice with custom skills and agentic coding workflows for AI-augmented development.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream after assessing that an LLM was not needed, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- **Tech stack: Python, LangChain, LangGraph, LiteLLM, LLM APIs, MCP, RAG, GraphRAG, Vector Search, Docker, Kubernetes, Azure, FastAPI, React, Neo4j, MongoDB, Postgres**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them, and translated the requirements into a solution approach.
- Built and deployed an agentic AI application with LangGraph that routes requests across vector database queries, general inference, and summarization, with a React frontend and a Python FastAPI backend.
- Containerized the stack with Docker and deployed it to AWS with CDK through GitLab CI/CD.
- **Tech stack: Python, LangChain, LangGraph, FastAPI, React, Docker, GitLab CI/CD, AWS (CDK, EC2, Lambda)**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated evaluation of answer quality and retrieval for the PrismaChat RAG system with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Built retrieval pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for better retrieval.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, Azure AI Search, AWS CDK**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, engineering lag features and rolling statistical windows from sensor data.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Spec-Driven Coding Agent Harness | Claude Code | FastMCP | Langfuse | CI/CD

- Harness that turns a plain-language feature request into a testable specification (acceptance criteria written as pytest tests) and lets a coding agent (Claude Code) implement it under a rule set: project conventions, custom skills, and hooks that block edits to protected files and secrets; an MCP server built with FastMCP gives the agent the spec and task context.
- A CI/CD pipeline gates every agent-written change with automated tests, static security analysis, and dependency scanning, plus an evaluation suite over benchmark tasks (pass rate, cost, turns per task); traces of the agent runs go to Langfuse for monitoring and observability.

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA

- Fine-tuned the open-source Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

TECHNICAL SKILLSET

- **Generative AI:** LLM APIs, prompt engineering, RAG architectures, LangChain, LangGraph, LiteLLM, agents and multi-agent systems, MCP servers, embedding models, GraphRAG, Hugging Face Transformers, fine-tuning (QLoRA)
- **LLM Evaluation:** RAGAS, PytrecEval, ground-truth evaluation of answer quality and retrieval
- **AI Development Tools:** Claude Code (custom skills, agentic coding workflows, MCP integration)
- **Programming:** Python, SQL, FastAPI, REST APIs, Python microservices, React, Streamlit
- **Vector Databases & Data:** Qdrant, Pinecone, Azure AI Search, Postgres, MongoDB, Neo4j, DynamoDB
- **Containers & CI/CD:** Docker, Kubernetes, GitLab CI/CD, Git, AWS (CDK, Lambda, EC2, SQS, S3), Azure
- **Machine Learning:** Scikit-Learn, XGBoost, PyTorch, TensorFlow, time-series analysis

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)